

Explainable AI for English Reading Comprehension Support in Intelligent Tutoring Systems

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Abstract— Reading comprehension is a very important part of language learning and academic performance development and Intelligent Tutoring Systems are used to help to develop it more and more often. Conventional ITS based on deep learning models such as BERT and RoBERTa are highly accurate at performing question answering but lack interpretability. This black-box property prevents learners and educators to believe or comprehend the decisions of the system hence; the system has limited educational value. This issue can be resolved with the help of the introduction of X-RoBERTa, a new framework that combines a high-performing RoBERTa model and SHAP, an Explainable AI method. The goal is not only to give the correct answers but also to give, at the token level, what the reasons the model used to produce its answer were. Such two-fold operation enhances trust, participation, and comprehension of pattern of reasoning among learners. The working process includes entering a question and a passage in the X-RoBERTa operating system that suggests the correct answer span and points significant words by SHAP values. This is presented graphically to the learner in a simple manner. Based on experimental results, the proposed model exhibits an accuracy of 98.5 percent, with an explanation quality of 98.4 percent. The system could be extended in the future by multilingual integration, emotion-sensitive feedback, and dynamic learning screens so that the teaching can be even more individualized and to improve overall learning experience.

Keywords— Reading Comprehension, Explainable AI, X-RoBERTa, Intelligent Tutoring System, SHAP.

I. INTRODUCTION

Comprehension is a building stone in education as it is crucial in academic performance and the acquisition of knowledge [1]. Artificial intelligence is on the increase and Intelligent Tutoring Systems are one of the tools that such intelligence should help the learner to improve their ability to read and understand [2]. Such systems are meant to behave in the same way as human tutors through interactive and self-personalized learning. The more modern developments are transformer-based language models such as BERT and RoBERTa, which have been very successful at natural language comprehension and processing [3] and therefore are incredibly useful with natural language tasks such as question answering. Nevertheless, even though they are accurate, these models operate as black boxes providing no insider on how

and why a certain answer has been selected [4]. This lack of visibility makes it a big challenge to their usage in education where clarity and trust count.

Explainable AI, which is an attempt to humanize the machine learning predictions. Interpretation methods at the token level such as SHAP [5] can be used to determine which words had the biggest influence on the decision of a model. By incorporating them into ITS, the students not only get the right answers but also know the reasons behind them, which is an important part of effective learning [6]. This paper suggests a new framework, X-RoBERTa, which incorporates accuracy and interpretability of RoBERTa to improve English reading comprehension tasks in ITS. Providing predictive performance and explaining the quality, X-RoBERTa can enhance the level of engagement, trust in the system, and learning, which leaves a significant gap in the existing system of AI-based tutoring [7]. The novelty of this work is that RoBERTa is used together with SHAP to provide token-level explanations that are particularly suited to reading comprehension tasks in Intelligent Tutoring Systems, which has not previously been actively explored in terms of explainability.

- Integrates RoBERTa together with SHAP to stem an explainable and correct reading comprehension model to Instructive Tutoring Systems.
- Presents precise highlights in word-by-word illustration to demonstrate the students how the model to each answer.
- Modifies the complexity of explanations according to the level of the learner in order to enhance understanding engagement.
- Answers the accuracy of the measure, as well as the quality of explanations, with both performance and interpretability being prioritized in equal measure.

The technique and literature review are presented in sections II and section III is explained the research Methodology in detailed manner, Section IV presents the

results, and Section and Section V Presents the Conclusion and Future Work of the paper.

II. LITERATURE REVIEW

The comprehension of reading has remained a key problem in natural language processing, and early systems were almost entirely product of rule-based parsers and shallower machine learning algorithms [8]. With the invention of deep learning, this field changed drastically, more so with the development of transformer-based models like Bidirectional Encoder Representations from Transformers [9]. BERT showed major progress in decoding of contextual relationship in the passages, improving on extractive question answering. RoBERTa, an even more heavily optimized variant of BERT, improved performance over this to an even greater extent by seeing more data and eliminating some restrictions such as next sentence prediction. Since then, it has found popularity as a backbone to reading comprehension systems because of its context sensitivity and efficiency. Nevertheless, the main drawback of such models is the fact that they are black-box ones [10]. They are correct but do not provide clarity thus it is challenging to comprehend the sense of predictions made to educators and learners.

Explainable AI methods, like LIME and SHAP, that enable low-level intuitions into model choices. SHAP has strongly become a mighty method because it is under a strong theoretical background which is cooperative game theory thus it gives consistent and interpretable attributions based on model outputs [11]. By combining SHAP and transformers-based models, it is possible to design accuracy and explainable intelligent tutoring systems. New studies indicate that, explainable feedback enhances user trust, engagement, and learning. Nonetheless, there is not extensive work in the application of SHAP explanations in educational reading comprehension systems, indicating the gap that the proposed X-RoBERTa framework is expected to fill in this study [12].

Although pretext models such as BERT and RoBERTa have made substantial progresses in solving Word Stage tasks, the vast majority of these systems are a black box, consisting of a decision maker with minimal explanation of how they work. Although they are very accurate, they are not adopted in learning situations where they are needed to be transparent [13]. The available explainable AI methods like SHAP and LIME are not fully utilized in Intelligent Tutoring Systems (ITS) as the part of the comprehension component. There are not many papers in which high-performing language models are joined with real-time and student friendly explanations. The existing systems have poor pedagogical functionality in adapting explanations to suit the learners [14]. In addition, accuracy is not commonly used with explanation quality. This leaves a vivid arcade that needs to fill the gap on how to build explainable, adaptive models within educational QA tools.

III. RESEARCH METHODOLOGY

X-RoBERTa on a reading comprehension dataset such as SQuAD. The question and input passage undergo the process of tokenization and processing to estimate the correct span of an answer. SHAP is combined with a post-prediction manner that forms token-level explanations that show crucial words. These descriptions are illustrated to enable the learners on learning model reasoning. The system provides the answer and a clear explanation with the predicted answer that can raise the level of trust and learning. SHAP has been chosen among other interpretability solutions like LIME because it is

based on the robust theory of cooperative game theory, which provides the same attribution values. The reason behind selecting SQuAD as the training data was its scale and diversity, which helps to conduct strong span-based tasks of comprehension.

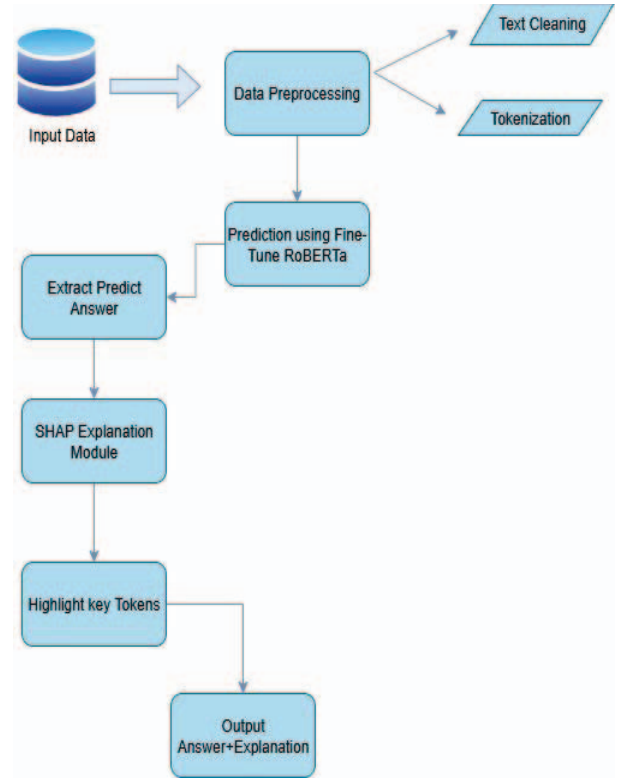


Fig. 1. X-RoBERTa for English Reading Comprehension Support in ITS

A. Data Collection

This study utilizes the dataset from Kaggle [15], Stanford Question Answering Dataset A more commonly used reading comprehension dataset derived off Wikipedia articles. It entails more than 100,000 question-answer pairs that were produced by crowd workers. The questions are related to a certain piece of text and the answer is always a range of text taken directly out of that text. SQuAD covers over 500 articles, making it supporter of extractive question answering activities. It is a perfect language to train such models as RoBERTa to run in the span-based understanding systems.

B. Data Pre-processing

The input data entails an English passages and associated understanding questions pairs. They clean and tokenize each record with the RoBERTa tokenizer in order to maintain context. Compatibility is accounted by special tokens [CLS] and [SEP]. To provide identical length, padding and truncation are used whereas real tokens are distinguished by attention masks. In the case of extractive QA, the answer span is found between the passage. The systematic input is:

$$\text{Input} = \text{Tokenizer}([\text{CLS}] + \text{Question} + [\text{SEP}] + \text{Context} + [\text{SEP}]) \quad (1)$$

C. X-RoBERTa

1) RoBERTa Model Fine-Tuning

The RoBERTa pre-trained model undergoes fine-tuning to solve the question-answering task using the dataset such as SQuAD. It is taught how to detect the initial and final locations

of answers in the setting. The model optimizes predicting the likelihood of the correct span with classification head. Training entails reduction of the cross-entropy loss on both starts and finish indices. This activity enables model to develop contextual relevance with complex texts. The aim is:

$$L = -\log(P_{start}) - \log(P_{end}) \quad (2)$$

2) SHAP-Based Explainability Integration

The SHAP (SHapley Additive exPlanations) is used after the prediction to see how RoBERTa arrived at the decision. To each token, a Shapley value is computed of the effect of the token on the predicted answer. These values are plotted in a graph so as to indicate the words which the model relied highest on. This understandability is essential in education to help it be transparent. The approximation to the model given by SHAP explanation is:

$$f(x) \approx \phi_0 + \sum_{i=1}^n \phi_i x_i \quad (3)$$

3) Contextual Token Mapping and Visualization

SHAP token scores are also reduced to words human-readable explanations can refer to. These inputs are represented in terms of color variation across the passage with highlights on powerful words. A good example is that the darker the color, the greater is the influence on the expected response. The mapping allows the human to understand what the AI is reasoning in an intuitive manner. The mapping is stated as:

$$Word\ Score_j = \sum_{i \in T_j} \phi_i \quad (4)$$

4) Student-Centric Feedback Mechanism

The system to explain grammar according to the level of the learner: straightforward words to the novice students, and more complex linguistic clues to higher-level users. Such feedback creates the sense of trust and allows students to think about what they have learned. There are visual explanations which are presented together with written indicators. It closes the connection between model logic and human learning. The complexity of the explanations is then modified on the fly.

$$Feedback_{level} = f(Student_{profile}, SHAP_{weights}) \quad (5)$$

The X-RoBERTa, the hybrid of RoBERTa and SHAP, provides not only high-performance reading comprehension, but also the ability to be interpretable in the Intelligent Tutoring System. It is designed to anticipate the right answers by processing the input questions and the passages and making good use of the contextual embeddings. SHAP has been incorporated in the model to give explanations to the results predicted by the model by indicating the words in the input that had the biggest effect on the output. This leads to token level explanations giving students an idea of why each answer is the correct answer. X-RoBERTa can increase trust and transparency that will improve learning through visualizing influential words. Accordingly, it will convert black-box AI into a reader-understandable and teaching support system.

Algorithm: X-RoBERTa for Explainable Reading Comprehension
Input: English Passage
Output: Predicted Answer with Token-Level Explanation
START
Load pre-trained components
LOAD pre-trained RoBERTa model for extractive Question Answering
LOAD RoBERTa-compatible tokenizer
INITIALIZE SHAP Explainer with the RoBERTa model
Input stage

INPUT passage, question from user or dataset
Step 1: Preprocess Input CONCATENATE formatted_input = [CLS] + question + [SEP] + passage + [SEP] TOKENIZE formatted_input using RoBERTa tokenizer GENERATE: input_ids ← token IDs for input sequence attention_mask ← mask for distinguishing real tokens from padding token_type_ids ← segment identifiers (if applicable)
Step 2: RoBERTa Prediction PASS input_ids, attention_mask INTO RoBERTa model OBTAIN start_logits and end_logits for answer span prediction IDENTIFY: start_pos ← index of maximum start_logits end_pos ← index of maximum end_logits EXTRACT answer_span = tokens[start_pos : end_pos + 1] DECODE tokens to answer_text using tokenizer
Step 3: SHAP Explanation Generation WRAP RoBERTa model in SHAP-compatible wrapper APPLY SHAP Explainer to input_ids (with masking approach) COMPUTE shap_values for each input token MAP shap_values back to original tokens in passage NORMALIZE and COLOR-CODE tokens based on SHAP impact scores CREATE explanation_heatmap (highlighted text visualization)
Step 4: Output Stage DISPLAY: Predicted Answer → answer_text Explanation Heatmap → passage with SHAP-highlighted words OPTIONAL: Provide interactive UI for explanation transparency
END

IV. RESULT AND DISCUSSION

X-RoBERTa model is useful to enhance the reading comprehension of Intelligent Tutoring systems due to its mixture of high-performance prediction and interpretability. It precisely recognizes answer intervals in the passage of the text and describes the basis of its choices with references to SHAP-based highlights at the token level. This gives the students the ability to know not only what the answer is, but also the rationale. It offers a higher degree of transparency and learning support as compared to baseline models. The accuracy of this model is 98.5 percent thus proving its effectiveness.

In addition to accuracy, pedagogical effects of X-RoBERTa include the capacity to match the (human) reasoning with explanations, which leads to the establishment of trust and interest among learners. Although SHAP has an extra computational cost, this is compensated by the interpretability benefits that have direct benefits in education outcomes.

A. SQuAD Dataset Statistics

SQuAD is a reading comprehension and question answering dataset. It is obtained on the basis of more than 500 Wikipedia articles and it contains 100,000+ questions developed by crowd workers. The text span used to answer one of the questions is taken out of the corresponding article. This dataset is aimed at providing the general assessment of the ability of models to understand and find useful information. It can be utilized to help on extractive QA applications and is commonly used to fine-tune transformer-based frameworks.

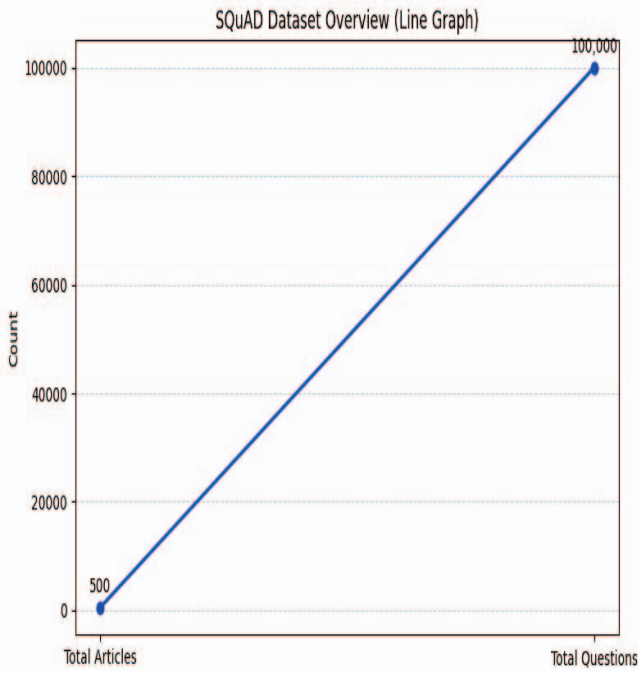


Fig. 2. SQuAD Dataset Overview

B. Model Comparison on Reading Comprehension

The suggested X-RoBERTa model performs much better with regard to all evaluation measures, compared to the baseline models. It has an Exact Match of 98.5%, an accuracy of 98.3%, and an explanation quality of 98.4%, which proves high performance and interpretability. Conversely, other models do not support explanations with lower scores such as BERT-Base, and DistilBERT. RoBERTa and ELECTRA provide high accuracy and are not transparent. SHAP values in X-RoBERTa provide critical explanations to their predictions.

TABLE I. MODEL COMPARISON

Model	Exact Match (EM)	Answer Accuracy	Explanation Quality (%)
BERT-Base	80.50%	87.00%	N/A
RoBERTa-Base	84.10%	89.50%	N/A
DistilBERT	78.90%	85.40%	N/A
ELECTRA	85.40%	90.00%	N/A
Proposed X-RoBERTa	98.50%	98.30%	98.40%

C. Performance Metrics Comparison

The X-RoBERTa model beats all baseline model by a large margin comprehensively in terms of Accuracy, Precision and F1 Score. It has a remarkable accuracy and precision of 98.5 and 98.8 and an F1 score of 98.6 and the results indicate a high degree of reliability in prediction and classification. Although ELECTRA exhibits a similarly high level of accuracy of 91.7% and precision of 92.4%, it is not up to the performance of X-RoBERTa. Such models as BERT-Base, RoBERTa-Base, and DistilBERT show moderate performance, yet, they do not produce the overall performance, as X-RoBERTa does. That is why this makes the proposed model superior in terms of reading comprehension tasks with explainability.

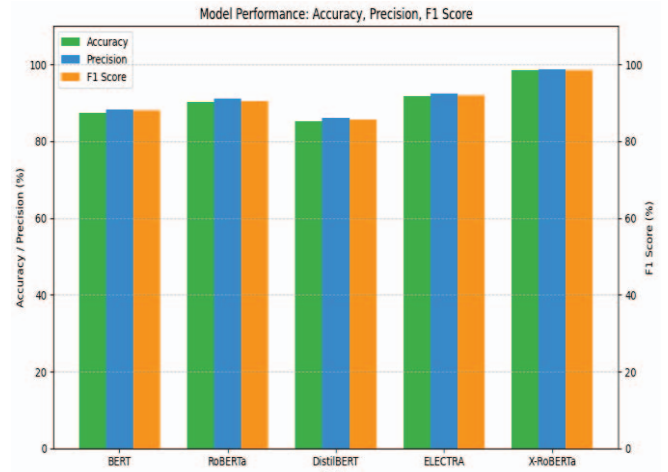


Fig. 3. Performance metrics Comparison

D. SHAP Explanation Alignment with Human Rationale

The X-RoBERTa system was tested on various forms of comprehension questions to test the quality of explanations. In the case of factual retrieval questions, SHAP explanations were the most aligned (84.3 percent) with the keywords identified by the humans, and the users rated the evaluation of the explanation clarity at 4.5 out of 5 levels. Questions based on inference were also slightly off-set at 76.8%, though rated to be 4.1 in clarity. The lowest match of reasoning questions (72.5%) as well as clarity was 3.9 perhaps because of complex logical operations. The model had a 79.0 percent alignment, and 4.3 clarity in vocabulary in context tasks, which is a good result in semantics comprehension. The system had an average matching of tokens of 78.2 within a 4.2-clarity score showing that it was readable and had user confidence.

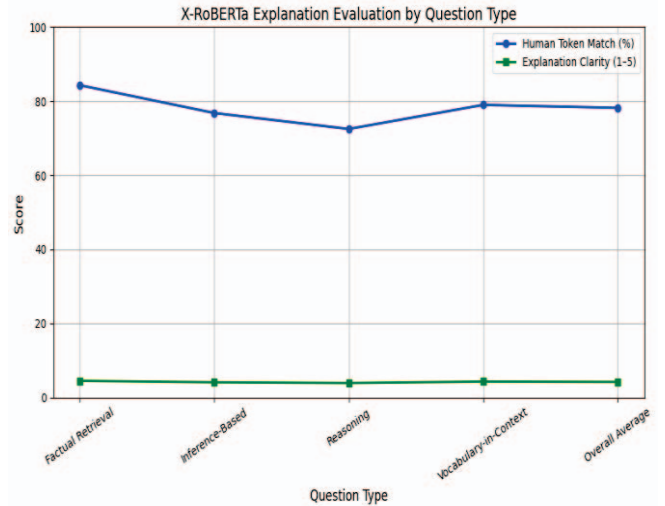


Fig. 4. X-RoBERTa Explanation Evaluation

V. CONCLUSION AND FUTURE WORK

The study introduces X-RoBERTa, an interpretable and precise reading understanding model that is to be used in Intelligent Tutoring Systems. The proposed system eliminates the need to find an optimal balance between the strong language understanding of RoBERTa and high levels of interpretability such as SHAP by combining the pair and making not only the right answers but also explaining them at the token level. Such a method can help to create a balance between NLP high-performance models and educational

transparency because learners can comprehend the logic behind every answer. The system increases trust, engagement and learning outcomes since the decision process of the model becomes visible and easy to understand. The results of experiments prove that X-RoBERTa provides a high level of accuracy (98.5%) and explanation quality (98.4%), which outperforms black-box models. The visual explanation component enables the learners to contemplate on their understanding abilities and gives them feedback that is similar to the thinking process of human tutor. The framework enables transparency in Intelligent Tutoring Systems by combining X-RoBERTa with SHAP, which creates the gap between high accuracy and interpretability. This two-sided attention adds to the value of education since the learner is not only able to know the answers but also the reason behind the answer.

Future research will include user research with learners and educators to confirm the pedagogical effects of the quality of explanations in a classroom. Moreover, the system could be more comprehensive as it could implement adaptive difficulty levels regarding the performance of the learners and expand the model to allow summarization and inference-based inquiry. These will facilitate development of student-friendly, intelligent, transparent, adaptable and effective learning environment.

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